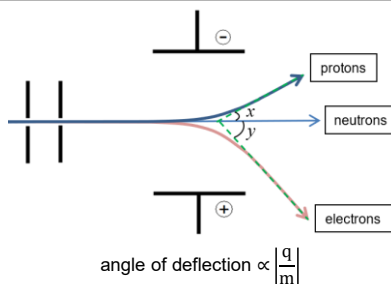


Deflection of particles



In a particular experimental set-up, a beam of He^{2+} particles are deflected in an electric field by an angle of $+10^\circ$. What angles will the beam of each of the following particles be deflected?

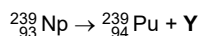
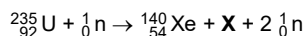
Li^+	
N^{3-}	

Proton no. & Nucleon no.

Subatomic particles

proton	relative mass: 1 relative charge: +1	1p
neutron	relative mass: 1 relative charge: 0	1n
electron	relative mass: $\frac{1}{1840} \approx 0$ relative charge: -1	0e

Deduce the identity of X and Y



Orbital shapes

s - spherical - non-directional	1s	2s
p - dumb-bell - directional	2p _x	2p _y
	2p _z	3p _z
d	3d _{xy}	3d _{yz}
	3d _{xz}	3d _{x²-y²}
	3d _{z²}	

Electronic configuration, Electron-in-box & Energy level

Method

- order of filling up electrons:
 $1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow \dots \rightarrow 4p \rightarrow 5s$
- order of writing electronic configuration:
 $1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow \dots \rightarrow 4p \rightarrow 5s$
- orbitals can hold a maximum of two electrons and they must be of opposite spins.
- orbitals must be occupied singly before pairing.
- both Cr and Cu have anomalous electronic configuration.

Representations for Na:

Na							
1s	2s	2p			3s		

Energy ↑

- State the electronic configuration of the following species

Fe	
Fe ⁺	
Fe ³⁺	

Note: For writing of electronic configuration of ions, always start from the atom

Explaining Trends

Key points to explain atomic trends:

- nuclear charge
- shielding effect (no. of e⁻ shell & no. of e⁻)
- electrostatic attraction between nucleus and outermost e⁻

atomic radii	- down the group: _____ - across the period: _____
ionic radii	- down the group: _____ - across the period: - decreases across cations - sharp increase from cation to anion - decrease across anions
electro-negativity	- down the group: _____ - across the period: _____ <i>Note:</i> electrostatic attraction between nucleus and bonding e⁻

1st ionisation energy

- down the group: _____
- across the period: _____

exception #1: ns² vs ns² np¹

s	p
s	p

p orbital is _____ in energy level compared to s orbital, hence less energy required to remove the electron in p orbital

exception #2: ns² np³ vs ns² np⁴

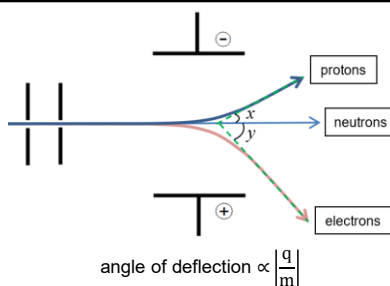
s	p
s	p

_____ electron in p orbital experiences additional _____ repulsion, hence less energy is required to remove the electron

successive I.E.

- _____

Deflection of particles



In a particular experimental set-up, a beam of He^{2+} particles are deflected in an electric field by an angle of $+10^\circ$. What angles will the beam of each of the following particles be deflected?

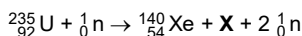
Li^+	For He^{2+} : $+10^\circ = k \left(\frac{+2}{4}\right)$ $k = 20$ For Li^+ : angle = $20 \left(\frac{+1}{4}\right)$ $= +2.86^\circ$
N^{3-}	For N^{3-} : angle = $20 \left(\frac{-3}{14}\right)$ $= -4.29^\circ$

Proton no. & Nucleon no.

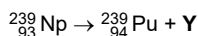
Subatomic particles

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neutron	relative mass: 1 relative charge: 0	1n
electron	relative mass: $\frac{1}{1840} \approx 0$ relative charge: -1	0e

Deduce the identity of X and Y



$$\begin{aligned} \text{a X} \quad 235 + 1 &= 140 + a + 2(1) \Rightarrow a = 94 \\ \text{b} \quad 92 + 0 &= 54 + b + 2(0) \Rightarrow b = 38 \end{aligned}$$



$$\begin{aligned} \text{a Y} \quad 239 &= 239 + a \Rightarrow a = 0 \\ \text{b} \quad 93 &= 94 + b \Rightarrow b = -1 \end{aligned}$$



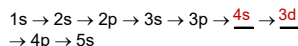
Orbital shapes

s - spherical - non-directional		
p - dumb-bell - directional		
d		

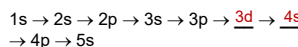
Electronic configuration, Electron-in-box & Energy level

Method

- order of filling up electrons:



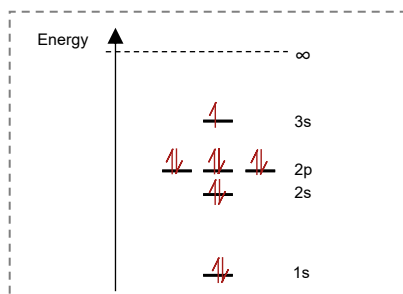
- order of writing electronic configuration:



- orbitals can hold a maximum of two electrons and they must be of opposite spins.
- orbitals must be occupied singly before pairing.
- both Cr and Cu have anomalous electronic configuration.

Representations for Na:

Na	$1s^2 2s^2 2p^6 3s^1$						
							
1s	2s	2p			3s		



- State the electronic configuration of the following species

Fe	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
Fe^+	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^1$
Fe^{3+}	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

Note: For writing of electronic configuration of ions, always start from the atom

Explaining Trends

Key points to explain atomic trends:

- nuclear charge
- shielding effect (no. of e^- shell & no. of e^-)
- electrostatic attraction between nucleus and outermost e^-

atomic radii	- down the group: <u>increases</u> - across the period: <u>decreases</u>
ionic radii	- down the group: <u>increases</u> - across the period: - decreases across cations - sharp increase from cation to anion - decrease across anions
electro-negativity	- down the group: <u>decreases</u> - across the period: <u>increases</u> Note: electrostatic attraction between nucleus and bonding e^-

1st ionisation energy

- down the group: decreases
- across the period: increases

exception #1: ns^2 vs $ns^2 np^1$

s	p
s	p

p orbital is higher in energy level compared to s orbital, hence less energy required to remove the electron in p orbital

exception #2: $ns^2 np^3$ vs $ns^2 np^4$

s	p
s	p

paired electron in p orbital experiences additional electron-electron repulsion, hence less energy is required to remove the electron

successive I.E.

- increases